

e searched, Not

Classification Metrics

Classification metrics tells us how good a classification model is.

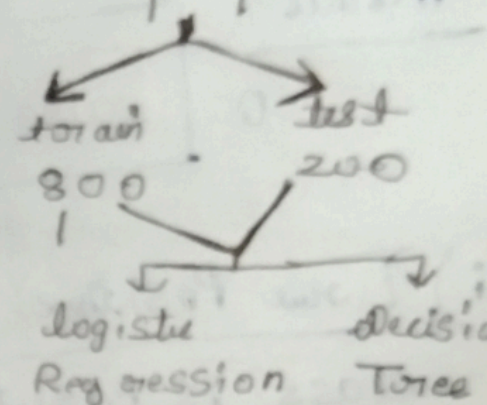
Accuracy:- Overall Correctness of the model.

Logistic Prediction Decision Tree Prediction

1000 data

actual label	Logistic Prediction	Decision Tree Prediction
1	✓ 1	✓ 1
0	✗ 1	✗ 1
0	✓ 0	✓ 0
0	✓ 0	✓ 0
0	✓ 0	✓ 0
1	✗ 1	✓ 0
1	✓ 1	✓ 1
1	✗ 0	✗ 0
0	✗ 1	✗ 1
0	✓ 0	✗ 1

Cgpa	iq	Placement
-	-	-
-	-	-



$$\text{accuracy} = \frac{\text{no. of } \checkmark}{\text{total prediction}} = \frac{8}{10} = 0.8 \Rightarrow 80\%$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Ex:- If 90 out of 100 predictions are correct:
accuracy = 90%

Problem with accuracy:- Fails on imbalanced datasets.

Ex:- 95 healthy, 5 disease

• model predicts all healthy $\rightarrow$ 95% accuracy but useless

How much accuracy is good:-

accuracy is Problem dependent.

- | | acceptable accuracy |
|--|--|
| $\rightarrow$ spam detection | $\rightarrow$ 90% + |
| $\rightarrow$ Medical diagnosis | $\rightarrow$ Even 95% may not be enough |
| $\rightarrow$ Recommendation Systems | $\rightarrow$ 70-80% can be useful |
| $\rightarrow$ Credit default | $\rightarrow$ depends on recall & risk |
| $\rightarrow$ accuracy is meaningful only when compared to | |
| • Baseline model | • Business requirement |

Problems:-

- ii) ignores type of error
- ii) doesn't show class-wise performance $\rightarrow$ doesn't tell which class is failing

High accuracy $\neq$ good model

ii) Confusion Matrix:-

		Predicted	
		1	0
Actual	1	True Positive [TP]	False ^{Negative} Negative [FP]
	0	False ^{Negative} Negative [FN]	True Negative [TN]

i) True Positive [TP]:- model predicts positive, actual class is positive. ☒ correct prediction

Ex:- Patient has cancer $\rightarrow$ model predicts cancer

→ ii) False Positive (FP) :- model predicts positive, actual class is Negative. ☒ wrong prediction, (Type I error)
Ex:- healthy patient → model predicts cancer

ii) False Negative (FN) :- model predicts negative, actual class is positive. ☒ wrong prediction (Type II error)
Ex:- Patient has cancer → model predicts healthy

→ most dangerous error

iv) True Negative (TN) :- model predicts Negative, actual class is negative. ☒ prediction
Ex:- healthy patient → model predicts healthy

Confusion matrix shows

- where the model is failing
- which type of error is high
- where FP or FN is more serious

iii) Precision :-

when my model predicts yes / positive, how often is it correct?

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Precision tells how reliable positive predictions are.
- High Precision $\Rightarrow$ fewer False positives, more trust in predictions.

When should focus on Precision :-

- Precisions are important when False Positives are costly.

Examples :-

(Ex:- 20LPA offer letter mail send as spam)

- Spam filter $\Rightarrow$ important email marked spam
 - loan approval $\Rightarrow$ bad customer approved
 - fraud alert $\Rightarrow$ genuine customer blocked
- } In these cases you want high Precision

→ Precision measures the proportion of correctly predicted positive instances among all predicted positive instances.

Ex:-

	Sent to Spam	Not sent to Spam
Spam	100 (TP)	170 (FN)
Not spam	30 (FP)	700 (TN)

	Sent to Spam	Not sent to Spam
Spam	100 (TP)	190 (FN)
Not spam	10 (FP)	700 (TN)

$$P_A = \frac{100}{100+30} \Rightarrow 0.75$$

$$P_B \Rightarrow \frac{100}{100+10} = 0.909$$

$$P_A < P_B$$

iv) Recall:-

out of all the Actual positive cases, how many did the model correctly find?

$$\text{Recall} = \frac{TP}{TP+FN}$$

→ Recalls tell you: how good the model is at not missing positives.

- How many important cases are caught vs missed.

High recall $\Rightarrow$ • fewer false Negatives

- Less chance of missing critical cases.

Recall is important when False Negatives are dangerous

Ex:- • Cancer detection $\rightarrow$ missing cancer

• Fraud detection $\rightarrow$ fraud not detected

Precision $\Rightarrow$ Was I correct when I said Yes?

Recall $\Rightarrow$ Did I catch all Yes cases?

→ Recall measures the proportion of actual positive instances that are correctly identified by the model.

High recall $\Rightarrow$ fewer missed important cases

	Detected Cancer	Not Detected
has cancer	1000 (TP)	200 (FN)
No Cancer	800 (FP)	800 (TN)

$$R_A = \text{accuracy} \Rightarrow 0.9$$

FN = 200 } less

so $\Rightarrow R_A$ is best //

$$R_A \Rightarrow \frac{1000}{1000 + 200} = \uparrow$$

	Detected Cancer	Not Detected
has cancer	1000 (FP)	500 (FN)
No Cancer	500 (FP)	800 (TN)

$$R_B = \text{accuracy} 0.9$$

FN $\Rightarrow$ 500 } more

$$R_B \Rightarrow \frac{1000}{1000 + 500} = \downarrow$$

V) F1 Score :-

F1 Score is a metric that balances Precision & Recall

$\rightarrow$ How good is the model when we care about both false positives and false negatives,

$$\boxed{F1 \text{ Score} = \frac{2 \times \text{Precision} \times \text{recall}}{\text{Precision} + \text{Recall}}}$$

Ex:-

$$\text{Precision} = 0.8$$

$$\text{recall} = 0.5$$

$$\Rightarrow F1 = \frac{2 \times 0.8 \times 0.5}{0.8 + 0.5} = \frac{0.8}{1.3} = 0.615 //$$

$\Rightarrow$ F1 is closer to the smaller value (recall)

$\rightarrow$ This penalizes the imbalance b/w precision and recall

$\rightarrow$ F1 Score is the harmonic mean of precision and recall, used to evaluate classification models on imbalanced datasets.

Precision - Recall Trade Off :-

You usually cannot increase Precision and Recall at the same time. Improving one often other worsens the other.

Why does this trade-off happen?
 Because models make decisions using a threshold
 (especially probabilistic models).

- default = 0.5
- probability $\geq 0.5 \Rightarrow$ Positive
- probability $< 0.5 \Rightarrow$ Negative

$\Rightarrow$ Changing this threshold causes the trade-off

Case 1: Increase Recall (catch more frauds) [aggressive catches everything]

- Lower the threshold:
- $\rightarrow$ more transactions labeled as fraud
- $\rightarrow$ fewer frauds missed (FN $\downarrow$)
- $\rightarrow$ But more normal transactions flagged (FPT)

result:-

- Recall $\uparrow$
- Precision $\downarrow$ (more false alarms)

Case 2: Increase Precision (reduce false alarms) [Conservative predicts only sure cases]

- Increase the threshold:
- $\rightarrow$ Only very confident frauds predicted.
- $\rightarrow$ Fewer false positives (FP $\downarrow$)
- $\rightarrow$ But some frauds missed (FNT)

result:-

- Recall $\downarrow$
- Precision $\uparrow$

MultiClass Precision:-

	<u>Predicted</u>			
	Dog	Cat	Rabbit	Total
Dog	25	5	10	40
Cat	0	30	4	34
Rabbit	4	10	20	
Total	29	45	34	

If there is huge difference used weighted

$$P_{\text{Dog}} = \frac{25}{29} = 0.86\%$$

$$P_{\text{Cat}} = \frac{30}{45} = 0.66\%$$

$$P_{\text{Rat}} = \frac{20}{34} = 0.58\%$$

In this case use avg macro